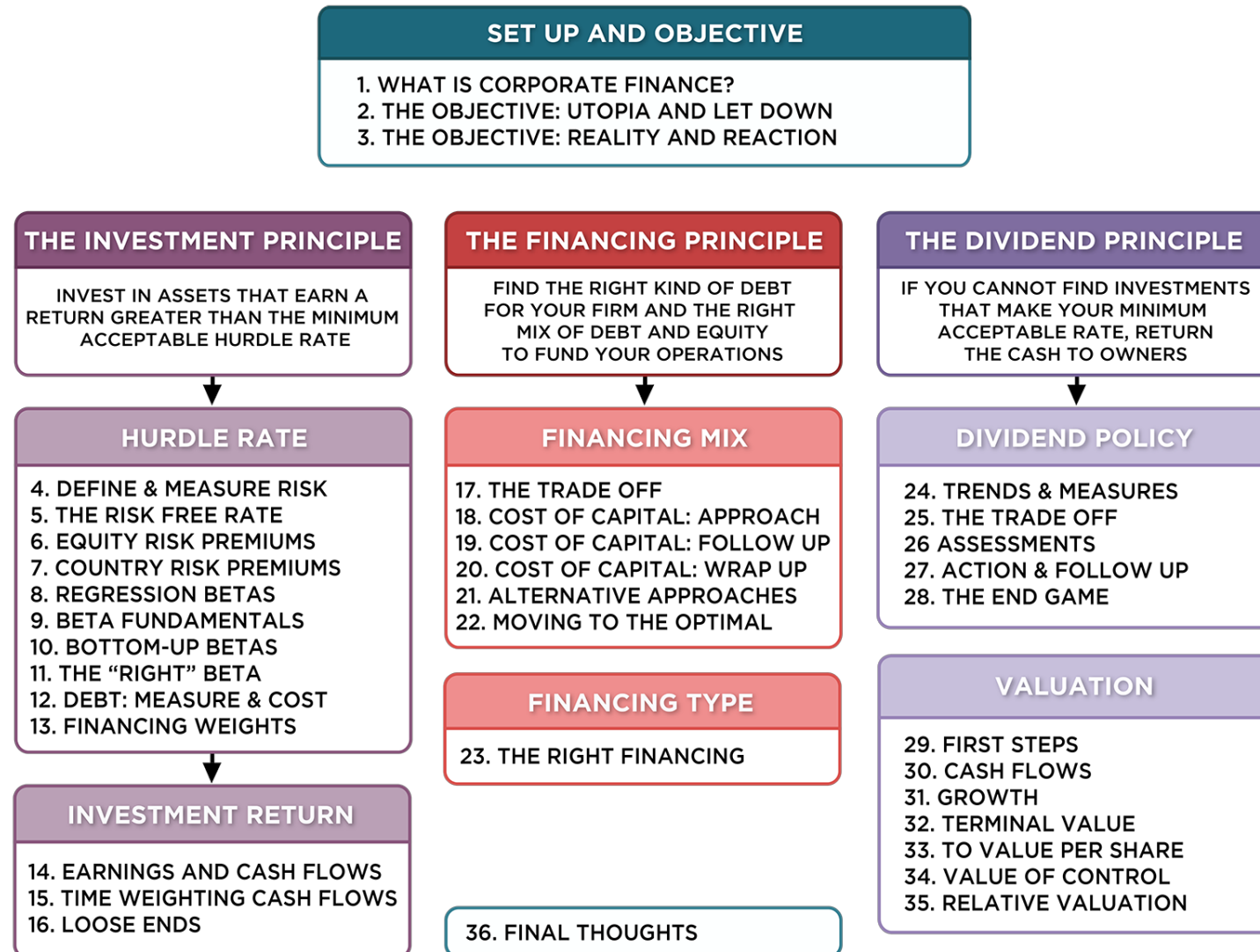


SESSION 10: BOTTOM-UP BETAS

If you cannot find comparable companies, it is because you have not looked hard enough.

SESSION 10: BOTTOM-UP BETAS



BETAS ARE WEIGHTED AVERAGES

- The beta of a portfolio is always the market-value weighted average of the betas of the individual investments in that portfolio.
- Thus,
 - the beta of a mutual fund is the weighted average of the betas of the stocks and other investment in that portfolio
 - the beta of a firm after a merger is the market-value weighted average of the betas of the companies involved in the merger.

BOTTOM-UP VERSUS TOP-DOWN BETA

- The top-down beta for a firm comes from a regression
- The bottom up beta can be estimated by doing the following:
 - Find out the businesses that a firm operates in
 - Find the unlevered betas of other firms in these businesses
 - Take a weighted (by sales or operating income) average of these unlevered betas
 - Lever up using the firm's debt/equity ratio
- The bottom up beta is a better estimate than the top down beta for the following reasons
 - The standard error of the beta estimate will be much lower
 - The betas can reflect the current (and even expected future) mix of businesses that the firm is in rather than the historical mix

DISNEY'S BUSINESSES: THE FINANCIAL BREAKDOWN (FROM 2013 ANNUAL REPORT)



2013 ANNUAL REPORT

<i>Business</i>	<i>Revenues</i>	<i>Operating Income</i>	<i>D&A</i>	<i>EBITDA</i>	<i>S, G & A Costs</i>	<i>Cap Ex</i>	<i>Identifiable Assets</i>
Media Networks	\$20,356	\$6,818	\$251	\$7,069	\$2,768	\$263	\$28,627
Parks & Resorts	\$14,087	\$2,220	\$1,370	\$3,590	\$1,960	\$2,110	\$22,056
Studio Entertainment	\$5,979	\$661	\$161	\$822	\$2,145	\$78	\$14,750
Consumer Products	\$3,555	\$1,112	\$146	\$1,258	\$731	\$45	\$7,506
Interactive	\$1,064	-\$87	\$44	-\$43	\$449	\$13	\$2,311



UNLEVERED BETAS FOR BUSINESSES

Unlevered Beta
(1 – Cash/Firm Value)



Business	Comparable Firms	Sample Size	Median Beta	Median D/E	Media Tax Rate	Company Unlevered Beta	Media Cash/Firm Value	Business Unlevered Beta
Media Networks	US firms in broadcasting business	26	1.43	71.09%	40.00%	1.0024	2.80%	1.0313
Parks & Resorts	Global firms in amusement park business	20	0.87	46.76%	35.67%	0.6677	4.95%	0.7024
Studio Entertainment	US movie firms	10	1.24	27.06%	40.00%	1.0668	2.96%	1.0993
Consumer Products	Global firms in toys/games production and retail	44	0.74	29.53%	25.00%	0.6034	10.64%	0.6752
Interactive	Global computer gaming firms	33	1.03	3.26%	34.55%	1.0085	17.25%	1.2187

A CLOSER LOOK AT THE PROCESS: STUDIO ENTERTAINMENT BETAS

Company Name	Levered Beta	Market Cap	Total Debt	Firm Value	Cash	Cash/Firm Value	Enterprise Value	Marginal Tax Rate	Gross D/E Ratio	Unlevered Beta	Pure Play Beta	EV/Sales
SFX Entertainment	1.12	728.80	\$98.89	\$837.69	\$143.60	17.14%	\$694.09	40.00%	13.39%	1.04	1.25	11.20
Mass Hysteria Entertainment	1.19	0.24	\$1.13	\$1.37	\$0.00	0.00%	\$1.37	40.00%	477.94%	.031	0.31	12.45
Medient Studios	.093	3.21	\$3.18	\$6.39	\$0.05	0.81%	\$6.34	40.00%	99.07%	0.58	0.59	1.21
POW! Entertainment	.094	3.97	\$0.34	\$4.31	\$0.43	9.85%	\$3.89	40.00%	8.65%	0.89	0.99	1.92
MGM Holdings	1.29	3631.70	\$142.16	\$3,773.86	\$140.70	3.73%	\$3,633.16	40.00%	3.91%	1.26	1.31	1.92
Lions Gate Entertainment	1.20	4719.60	\$1,283.20	\$6,002.80	\$67.20	1.12%	\$5,935.60	40.00%	27.19%	1.03	1.04	2.28
DreamWorks Animation	1.32	2730.00	\$348.30	\$3,078.30	\$156.40	5.08%	\$2,921.90	40.00%	12.76%	1.23	1.29	3.81
Twenty-First Century Fox	1.28	77743.50	\$20,943.00	\$98,686.50	\$6,681.00	6.77%	\$92,005.50	40.00%	26.94%	1.10	1.18	3.20
Independent Film Development	1.61	1.32	\$0.96	\$2.28	\$0.05	2.20%	\$2.23	40.00%	72.35%	1.12	1.15	3.37
Odyssey Pictures Corp	2.60	0.30	\$1.64	\$1.94	\$0.00	0.10%	\$1.94	40.00%	551.12%	0.60	0.60	2.90
Average	1.35					4.68%		40.00%	129.33%	0.92	0.97	4.43
Aggregate	1.35	\$89,572.64	\$22,822.82	\$112,395.45	\$7,189.43	6.40%	\$105,206.02	40.00%	25.48%	1.17	1.25	3.09
Median	1.24					2.96%		40.00%	27.06%	1.03	1.10	3.05

BACKING INTO A PURE PLAY BETA: STUDIO ENTERTAINMENT

The Median Movie Company

Movie Business	97.04	Beta (movies) = 1.0093	Debt	21.30	Beta (debt) = 0
Cash Business	2.96	Beta (cash) = 0.0000	Equity	78.70	Beta (equity) = 1.24
Movie Company	100.0	Beta (company) = 1.0668			

1. Start with the median regression beta (equity beta) of 1.24
 2. Unlever the beta, using the media gross D/E ratio of 27.06%
$$\text{Gross D/E Ratio} = 21.30/78.70 = 27.06\%$$
 3. Take out the cash effect, using the median cash/value of 2.96%
$$(0.0296)(0) + (1 + 0.0296)(\text{Beta of movie business}) = 1.0668$$
$$\text{Beta of movie business} = 1.0668/(1 - 0.0296) = 1.0993$$
- Alternatively, you could have used the net debt to equity ratio**
- $$\text{Net D/E ratio} = (21.30 - 2.96)/78.70 = 23.30\%$$
- $$\text{Unlevered beta for movies} = 1.24 / [1 + (1 - 0.4)(0.233)] = 1.0879$$

DISNEY'S UNLEVERED BETA: OPERATIONS & ENTIRE COMPANY

Business	Revenues	EV/Sales	Value of Business	Proportion of Disney	Unlevered Beta	Value	Proportion
Media Networks	\$20,356	3.27	\$66,580	49.27%	1.0313	\$66,579.81	49.27%
Parks & Resorts	\$14,087	3.24	\$45,683	33.81%	0.7024	\$45,682.80	33.81%
Studio Entertainment	\$5,979	3.05	\$18,234	13.49%	1.0993	\$18,234.27	13.49%
Consumer Products	\$3,555	0.83	\$2,952	2.18%	0.6752	\$2,951.50	2.18%
Interactive	\$1,064	1.58	\$1,684	1.25%	1.2187	\$1,683.72	1.25%
Disney Operations	\$45,041		\$135,132	100.00%	0.9239	\$135.132.11	

Disney has \$3.93 billion in cash, invested in close to riskless assets (with a beta of zero). You can compute an unlevered beta for Disney as a company (inclusive of cash):

$$\begin{aligned}
 \beta_{\text{Disney}} &= \beta_{\text{Operating Assets}} \frac{\text{Value}_{\text{Operating Assets}}}{(\text{Value}_{\text{Operating Assets}} + \text{Value}_{\text{Cash}})} + \beta_{\text{Cash}} \frac{\text{Value}_{\text{Cash}}}{(\text{Value}_{\text{Operating Assets}} + \text{Value}_{\text{Cash}})} \\
 &= 0.9239 \left(\frac{135,132}{(135,132 + 3,931)} \right) + 0.00 \left(\frac{3,931}{(135,132 + 3,931)} \right) = 0.8978
 \end{aligned}$$

THE LEVERED BETA: DISNEY AND ITS DIVISIONS

- To estimate the debt ratios for division, we allocate Disney's total debt (\$15,961 million) to its divisions based on identifiable assets.

Business	Identifiable Assets (2013)	Proportion of Debt	Value of Business	Allocated Debt	Estimated Equity	D/E Ratio
Media Networks	\$28,627	38.04%	\$66,580	\$6,072	\$60,508	10.03%
Parks & Resorts	\$22,056	29.31%	\$45,683	\$4,678	\$41,005	11.41%
Studio Entertainment	\$14,750	19.60%	\$18,234	\$3,129	\$3,129	20.71%
Consumer Products	\$7,506	9.97%	\$2,952	\$1,592	\$1,359	117.11%
Interactive	\$2,311	3.07%	\$1,684	\$490	\$1,194	41.07%
Disney	\$75,250	100.00%		\$15,961	\$121,878	13.10%

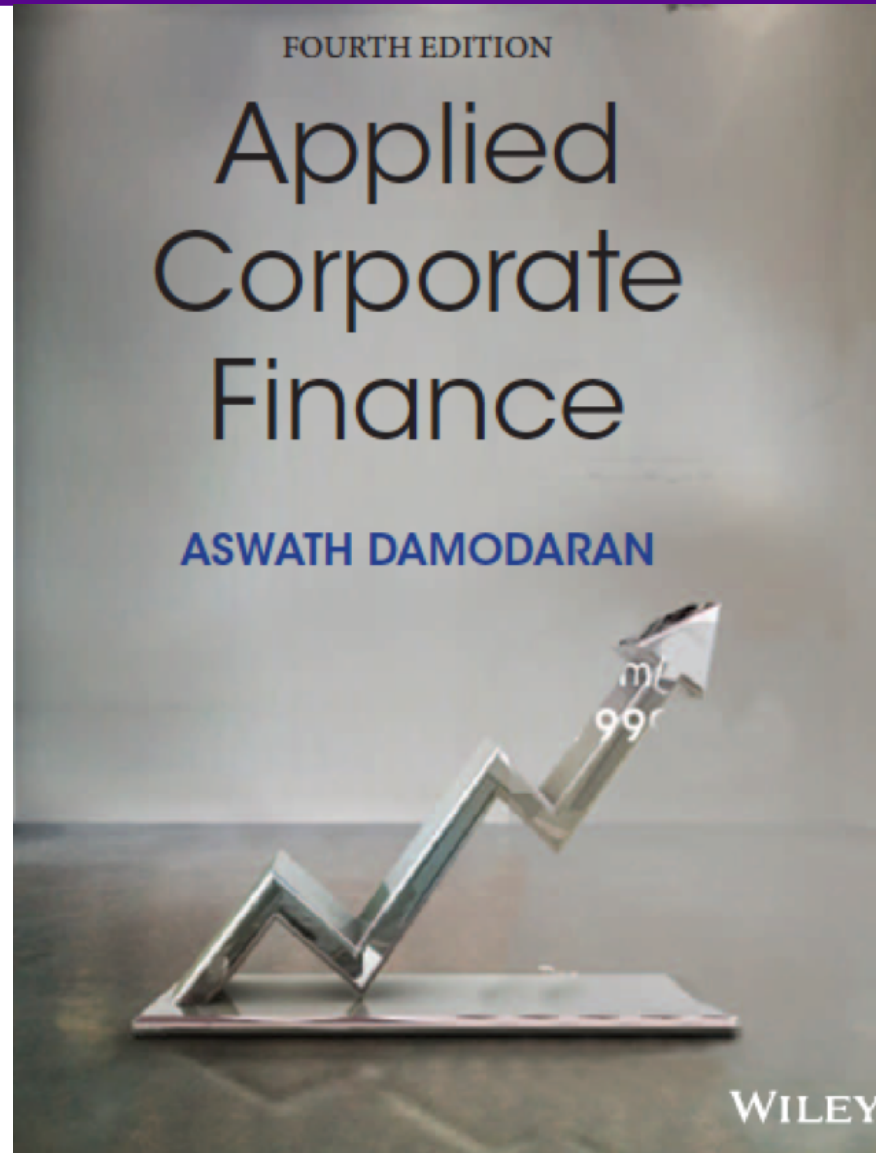
- We use the allocated debt to compute D/E ratios and levered betas.

Business	Unlevered Beta	Value of Business	D/E Ratio	Levered Beta	Cost of Equity
Media Networks	1.0313	\$66,580	10.03%	1.0975	9.07%
Parks & Resorts	0.7024	\$45,683	11.41%	0.7537	7.09%
Studio Entertainment	1.0993	\$18,234	20.71%	1.2448	9.92%
Consumer Products	0.6752	\$2,952	117.11%	1.1805	9.55%
Interactive	1.2187	\$1,684	41.07%	1.5385	11.61%
Disney	0.9239	\$135,132	13.10%	1.0012	8.52%

DISCUSSION ISSUE

- Assume now that you are the CFO of Disney. The head of the movie business has come to you with a new big budget movie that he would like you to fund. He claims that his analysis of the movie indicates that it will generate a return on equity of 9.5%. Would you fund it?
 - a. Yes. It is higher than the cost of equity for Disney as a company
 - b. No. It is lower than the cost of equity for the movie business.
- What are the broader implications of your choice?

Task
Estimate a
bottom-up
beta for your
company



Optional:
Read
Chapter 4